From Data to Decision: Applying WISCA for Empiric Antibiotic Coverage in Sepsis Using R

Presenters: Larisse Bolton[1,2], Matthijs Berends[3], Aislinn Cook[4]

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Introduction

Who and why?

To support **clinicians, researchers** and **public health experts** across the continent to pool resources for:

(1)**Development** and **implementation** of **data and specimen sharing** platforms

(2)**Research** and **clinical** collaboration

(3)**Capacity** development

Reduce neonatal sepsis burden

Learning outcomes

1. Explain the **purpose** and **function** of WISCA.
2. Describe **data structure requirements** for WISCA analysis.
3. **Prepare** and **clean** relevant sepsis datasets in R.
4. **Execute** a custom R function to generate WISCA outputs.
5. **Interpret** antibiotic coverage estimates to inform decision-making.

Presenters

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Department of Medical Microbiology and Infection Prevention, Faculty of Medical Sciences, University Medical Center Groningen, Netherlands

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Centre for neonatal and Paediatric Infection, City St. George's University London, UK.

Dr Larisse Bolton
South African Centre for Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, South Africa

Agenda

Time frame	Topic
10 minutes	Welcome & Orientation
20 minutes	WISCA 101
10 minutes	Data Requirements
25 minutes	Hands-On Part 1: Data Preparation in R
25 minutes	Hands-On Part 2: Running the WISCA Function
15 minutes	Interpreting outputs
10 minutes	Wrap-up: Integration and Limitations
5 minutes	Q & A

Training content

All training materials will be available from the Github repository located at https://github.com/BoltonL/WISCA_training.

The session will also be recorded, and the recording and slides will be made available to all participants.

WISCA 101

Empiric prescribing & guidelines

- Empiric prescribing is common particularly for neonatal and paediatric infections when the pathogen is still unknown
- Need to ensure patients are treated with best treatment choice while balancing antimicrobial stewardship needs
- Clinicians are treating patients with infections, not pathogens (frequently not known)
- Local microbiology data particularly where higher resistance rates to commonly recommended antibiotics is needed to inform empiric choices

Questions → how is best to incorporate local microbiology into informing empiric treatment choices and updating guidelines?

Traditional antibiogram

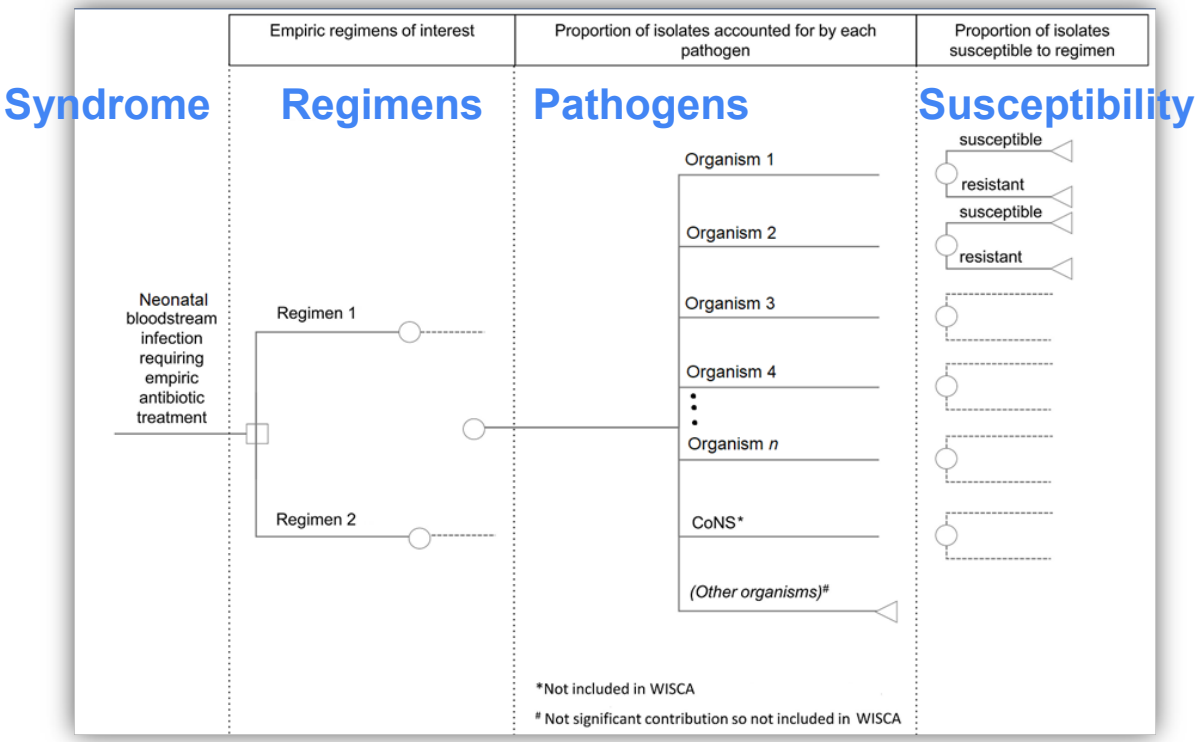
- Traditional antibiograms are limited in their clinical information
  - Usually provide information on number of bugs and number resistant to individual antibiotics & doesn't include combinations
- Clinicians are seeing patients with infection syndromes not about pathogens
- Many treatments are combination treatments which are not explicitly measured in traditional antibiograms
- Difficult to use antibiograms to select between alternative regimens
- Antimicrobial stewardship priorities
  - How to prioritise narrower spectrum regimens over broader spectrum regimens where appropriate?

WISCA

WISCA = Weighted Incidence Syndromic Combination Antibiogram

- **Weighted incidence** - accounts for relative distribution of pathogens AND their resistance patterns
  - More common bugs are weighted more than less common bugs
  - If high prevalence of resistance among small number of bugs, this is accounted for
- **Syndromic** - antibiogram assessed by patient syndrome (i.e. neonatal sepsis) and can be further stratified by risk groups within syndromes
- **Combination antibiogram** - explicitly looks at coverage of combination treatments (i.e. ampicillin + gentamicin) which are commonly used

Conceptualisation of a WISCA



Decision tree model

Inclusion of pathogens is local choice

*CoNS - only include when they are considered pathogens

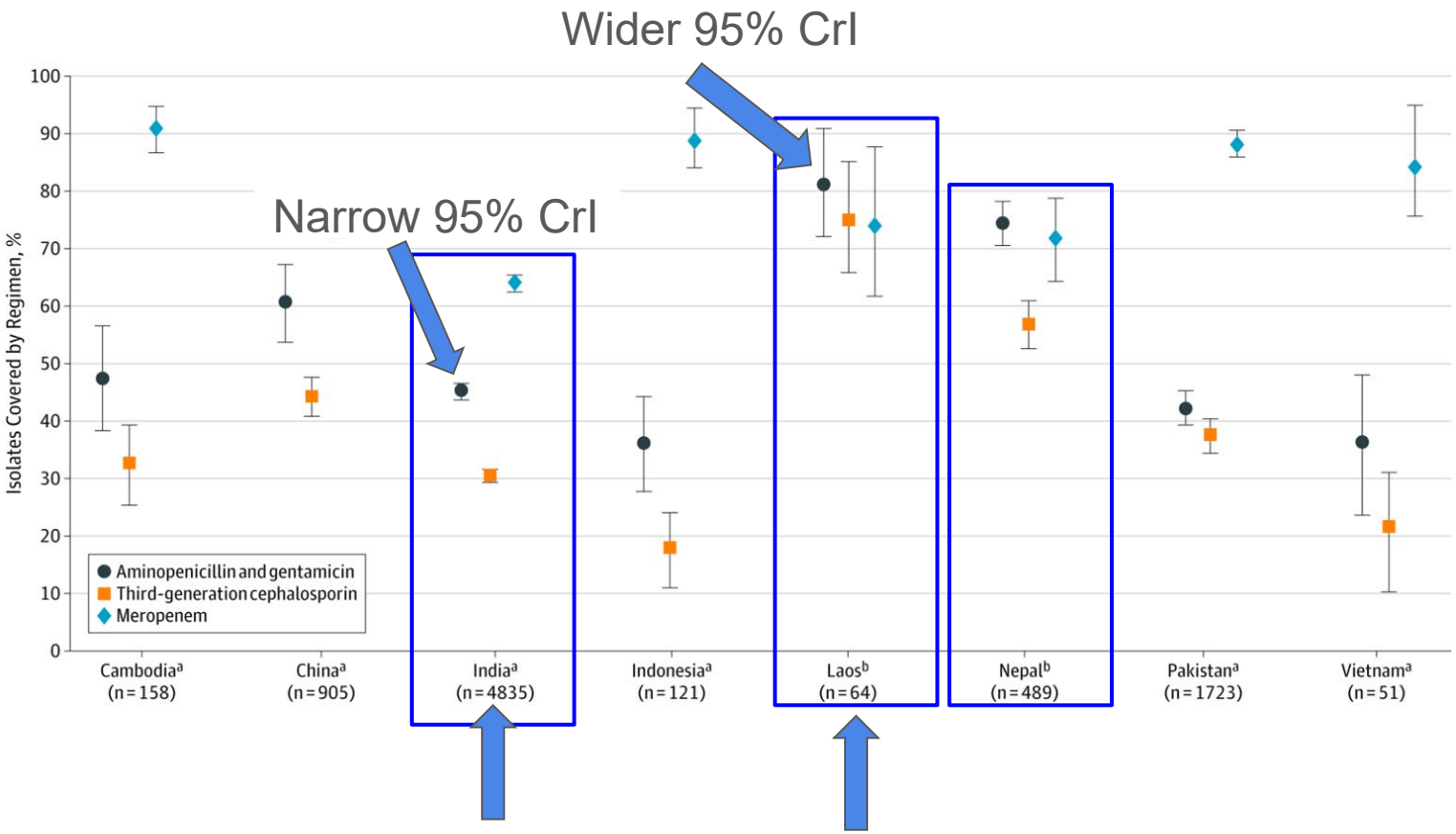
May want to only include most common pathogens i.e. 90% of pathogens

Coverage estimates

- Output of a WISCA are coverage estimates

- Coverage = Probability that that regimen would be “expected to be effective” for that clinical syndrome accounting for the most common pathogens and resistance patterns that cause that syndrome using the given data

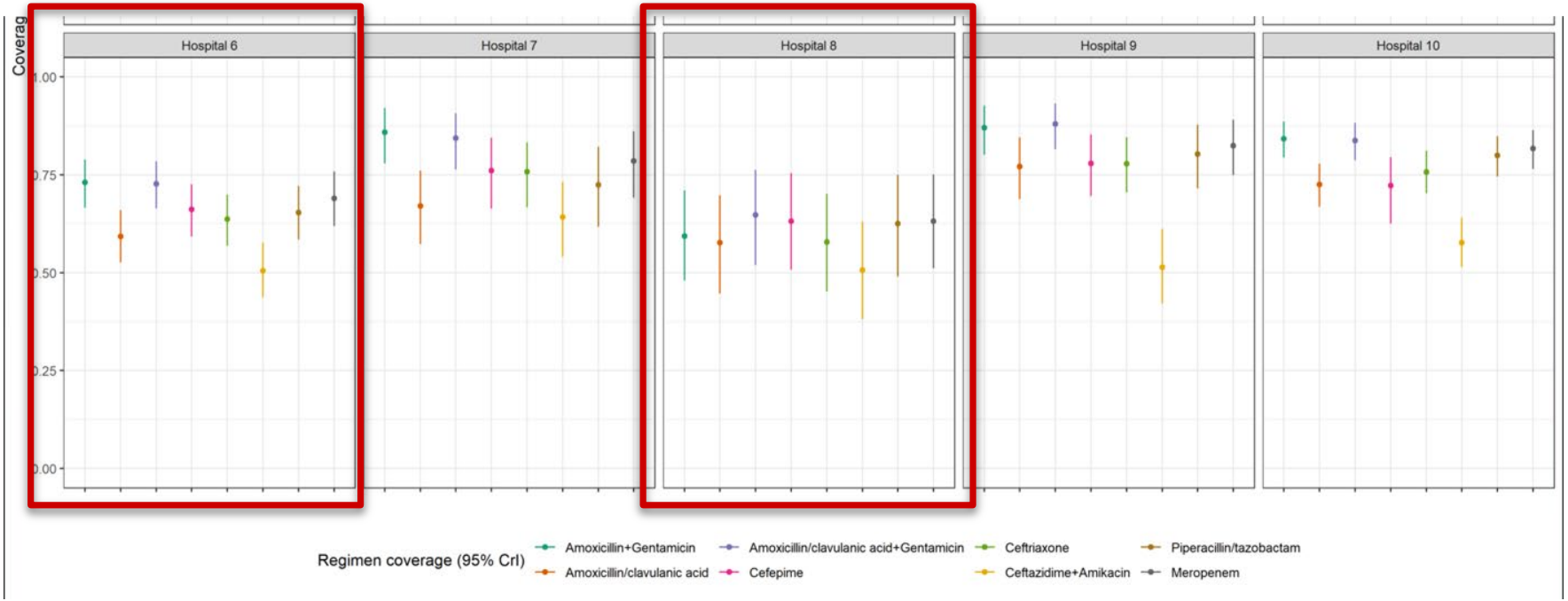
- Includes uncertainty intervals (95% credible intervals)
  - 95% CrI generated using Monte Carlo simulations
  - Accounts for imprecision due to small sample size
  - Or imprecision due to incomplete susceptibility testing
  - Allows for comparison between regimens - overlapping intervals = not statistically significant difference in coverage



Bielicki et al., 2020, PMID: 32049298

Hospital-wide

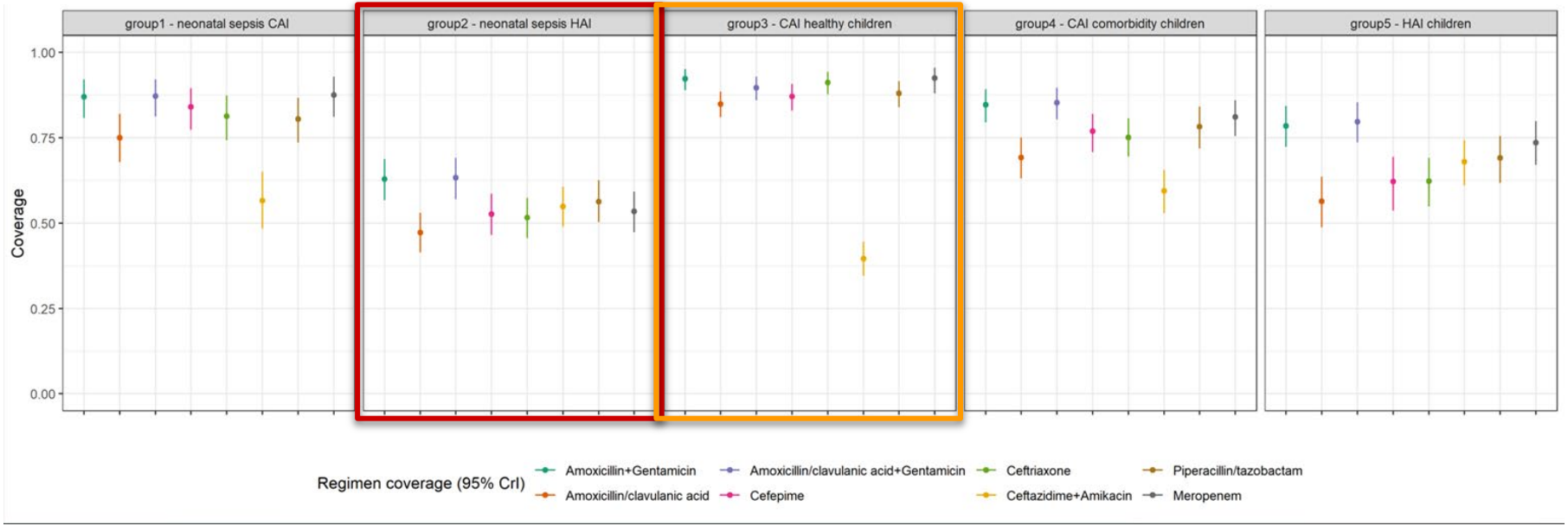
Stratification



Cook et al., 2023 PMID: 37252038

Syndrome stratification

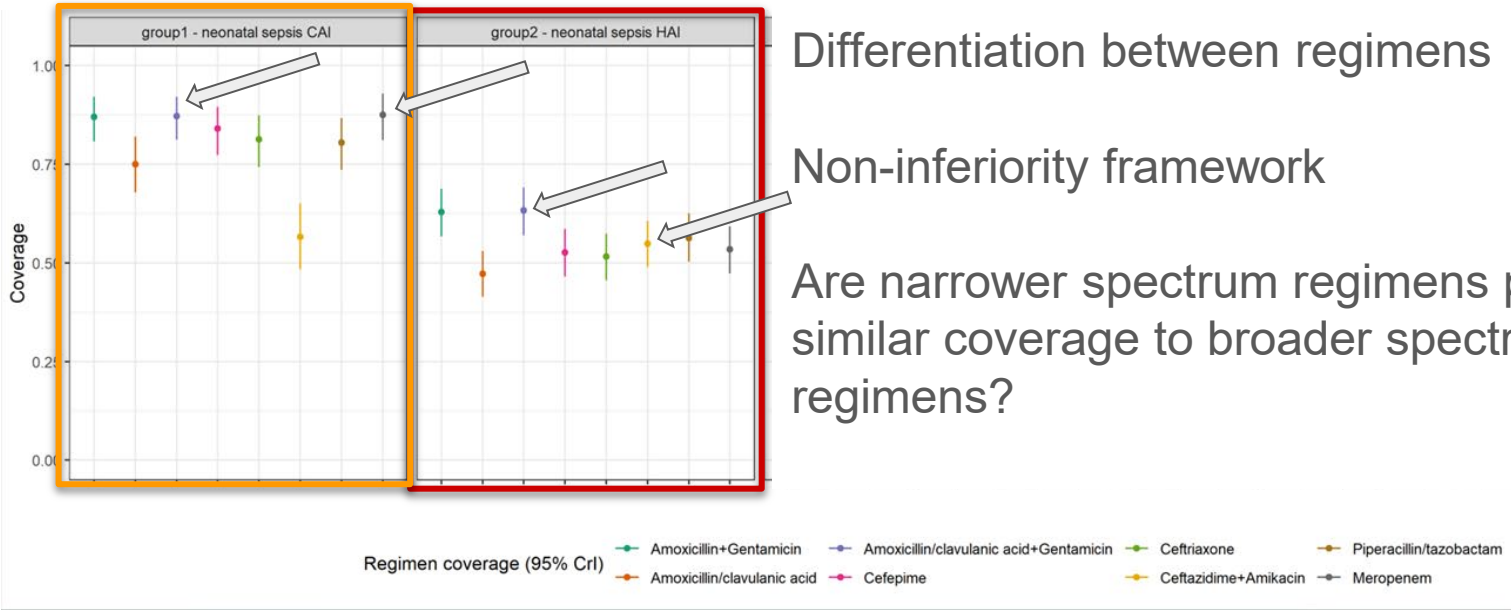
By clinical risk group



Important differences in coverage by clinical risk groups

Cook et al., 2023 PMID: 37252038

Comparison of regimens



Differentiation between regimens

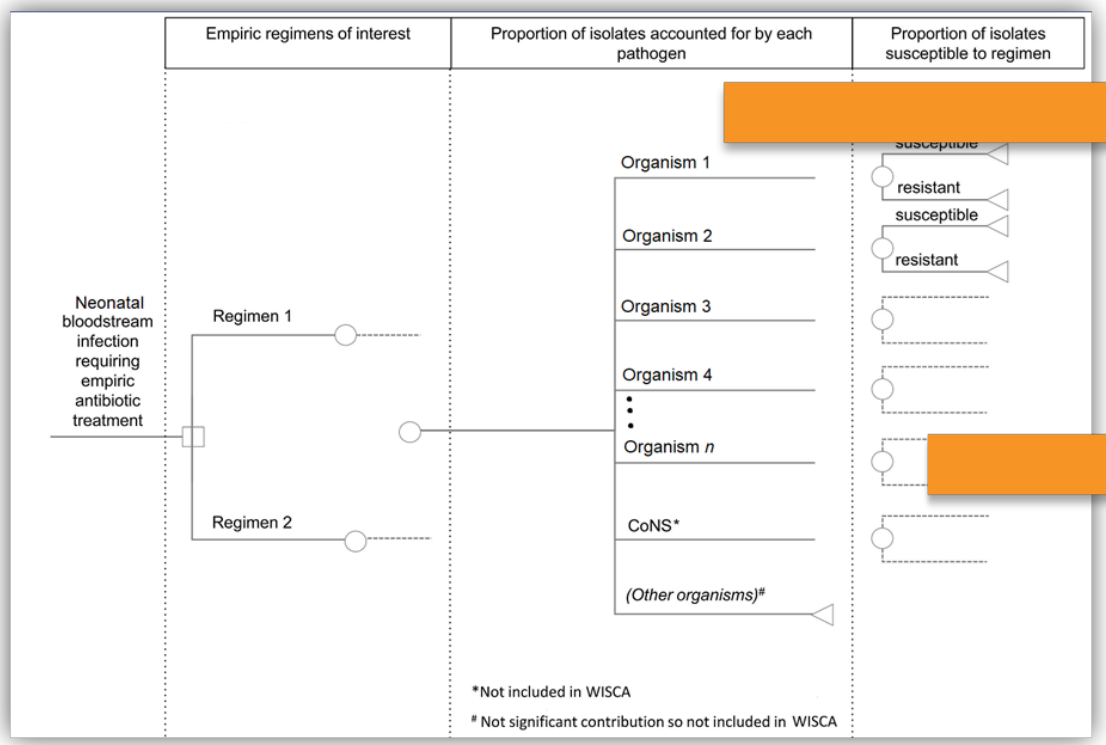
Non-inferiority framework

Are narrower spectrum regimens providing similar coverage to broader spectrum regimens?

Cook et al., 2023 PMID: 37252038

Data requirements

What does the WISCA need?



Incidence

Susceptibility

Adapted from Bielicki et al.(2016). Licensed under CC-BY-NC-ND

Dataset variables

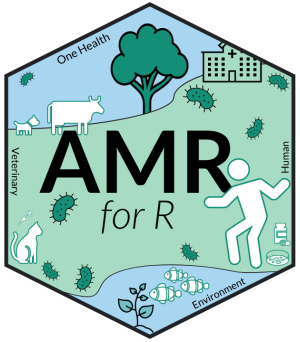
Essential variables:

- Date of specimen
- Unique patient identifier
- If more than 1 syndrome present in dataset, syndrome-identifying variable
- Antimicrobials of interest
- Microorganism names
- Antimicrobial susceptibility profiles

Additional variables (grouping):

- Level of analysis, e.g., ward/ hospital name
- Monomicrobial or polymicrobial classification
- Demographic variables e.g., sex or age
- Clinical variables e.g., outcome

Data considerations: **Naming**



Antimicrobials

ab	cid	name
AMP	6249	Ampicillin
SAM	119561	Ampicillin/sulbactam
AMR	73341	Amprolium
ANI	166548	Anidulafungin
APL	6602341	Apalcillin
APR	3081545	Apramycin
ARB	68682	Arbekacin
APX	71961	Aspoxicillin

Microorganisms

pathogen_name
Coagulase-negative Staphylococcus
Escherichia coli
Coagulase-negative Staphylococcus
Coagulase-negative Staphylococcus
Klebsiella pneumoniae

Data considerations: **Susceptibility**

> 30 isolates tested

- Complete **intrinsic** resistance for all organism/antibiotic combinations
- **Infer** susceptibility and resistance
- **Intermediate** susceptibility coding

Data considerations: **General**

- Any pathogens that need to be excluded?
- Species groupings? e.g., *Klebsiella spp.*
- Renaming/Regrouping of facilities/wards?
- All dates in same date format (e.g., year-month-day)
- Consisted coding of factors
- Erroneous data entries? e.g., Date of culture after date of outcome?
- Missingness in essential/needed variables?

Hands-On Part 1: Data Preparation in R

Unpacking preprocessing

- Read and display *wisca_dummy_messy.csv* in R
- Carefully evaluate the data
- What needs to be addressed in the data?

BloodCultureDate (character)	Patient ID (character)	Hospital (character)	Age at onset (character)	Sex (character)	GA_at_birth (double)	Birth wt (character)	Pathogen (character)	InfType (character)	Amp (character)	Gent (character)	Pip/Tazo (character)	Amk (character)	Mero (character)	Vanc (character)
18/09/2021	P0010	Chris Hani Baragwanath	1	M	32.6	1,919	Group B Streptococcus	EONS	S	R	S	Resistant	I	S
19/05/2021	P0126	Groote Schuur Hospital	6d	2	37.0	2190 g	Coagulase-negative Staphylococcus	HAI	R	I	I	Susceptible	S	S
19/07/2022	P0016	GSH	15	2	33.7	1130 g	Group B Streptococcus	HAI	R	S	Resistant	S	R	S
19/11/2021	P0065	Chris Hani Baragwanath	0d	M	32.3	1.83 kg	Coagulase-negative Staphylococcus	EONS	Resistant	S	Susceptible	S	I	S
2022-10-24	P0114	Tygerberg Hospital	2	female	32.4	1839	Acinetobacter baumannii	EONS	I	S	S	S	Susceptible	S
11/30/2022	P0124	Chris Hani Baragwanath	52d	2	34.2	2033 g	Group B Streptococcus	HAI	Susceptible	R	R	I	S	S
2021-12-20	P0114	Inkosi Albert Luthuli	96d	2	31.9	2222	Group B Streptococcus	HAI	R	R	S	S	S	S
06/21/2022	P0009	Groote Schuur Hospital	[illegible]	MALE	26.5	0.90 kg	Klebsiella pneumoniae	HAI	Resistant	S	S	S	S	I
05/12/2021	P0074	Groote Schuur Hospital	51	female	25.4	500 g	Klebsiella pneumoniae	HAI	Resistant	I	S	S	I	S
06/03/2022	P0001	Inkosi Luthuli	95	F	[illegible]	[illegible]	Staphylococcus aureus	HAI	R	S	I	R	Susceptible	I
02/22/2022	P0041	Inkosi Luthuli	41	1	29.1	1406	Acinetobacter baumannii	HAI	R	S	S	R	S	S
05/28/2021	P0116	Tygerberg Hospital	69	MALE	32.7	1994	Klebsiella pneumoniae	HAI	R	I	I	R	Susceptible	S
2022-06-22	P0017	Inkosi Luthuli	63	M	31.5	1867	Klebsiella pneumoniae	HAI	R	R	R	S	R	I
10/21/2022	P0101	Chris Hani Baragwanath	63	male	33.7	1956 g	Group B Streptococcus	HAI	I	S	S	S	S	Susceptible
09/19/2021	P0140	Universitas	1	MALE	31.2	1411	Coagulase-negative Staphylococcus	EONS	R	S	S	S	S	S
2022-05-12	P0055	GSH	67	M	34.0	1917 g	Escherichia coli	HAI	R	R	I	Susceptible	S	I
11/20/2021	P0089	Tygerberg Hosp	69d	2	33.2	1789 g	Klebsiella pneumoniae	HAI	R	I	I	R	R	S
2021-03-24	P0108	IALCH	11	male	35.9	2008 g	Staphylococcus aureus	HAI	S	S	R	S	Susceptible	Susceptible
05/04/2022	P0123	Chris Hani Baragwanath	80 wks	FEMALE	28.8	1371	Staphylococcus aureus	HAI	S	I	Resistant	Resistant	S	S
2022-07-12	P0063	GSH	57	1	25.3	854	Group B Streptococcus	HAI	S	Susceptible	S	I	S	S

Clean data

blood_culture_date (double)	patient_id (character)	hospital_name (character)	age_at_onset (double)	sex (character)	gestational_age_weeks (double)	birthweight_g (double)	pathogen_name (character)	infection_type (character)	ampicillin (character)	gentamicin (character)	piperacillin_tazobactam (character)	amika (cha
2021-09-18	P0010	Chris Hani Baragwanath	1	Male	32.6	1919	Group B Streptococcus	EONS	S	R	S	R
2021-05-19	P0126	Groote Schuur Hospital	6	Female	37.0	2190	Coagulase-negative Staphylococcus	HAI	R	I	I	S
2022-07-19	P0016	Groote Schuur Hospital	15	Female	33.7	1130	Group B Streptococcus	HAI	R	S	R	S
2021-11-19	P0065	Chris Hani Baragwanath	0	Male	32.3	1826	Coagulase-negative Staphylococcus	EONS	R	S	S	S
2022-10-24	P0114	Tygerberg Hospital	2	Female	32.4	1839	Acinetobacter baumannii	EONS	I	S	S	S
2022-11-30	P0124	Chris Hani Baragwanath	52	Female	34.2	2033	Group B Streptococcus	HAI	S	R	R	I
2021-12-20	P0114	Inkosi Albert Luthuli	96	Female	31.9	2222	Group B Streptococcus	HAI	R	R	S	S
2022-06-21	P0009	Groote Schuur Hospital	60	Male	26.5	901	Klebsiella pneumoniae	HAI	R	S	S	S
2021-12-05	P0074	Groote Schuur Hospital	51	Female	25.4	500	Klebsiella pneumoniae	HAI	R	I	S	S
2022-06-03	P0001	Inkosi Albert Luthuli	95	Female	29.5	1622	Staphylococcus aureus	HAI	R	S	I	R
2022-02-22	P0041	Inkosi Albert Luthuli	41	Male	29.1	1406	Acinetobacter baumannii	HAI	R	S	S	R
2021-05-28	P0116	Tygerberg Hospital	69	Male	32.7	1994	Klebsiella pneumoniae	HAI	R	I	I	R
2022-06-22	P0017	Inkosi Albert Luthuli	63	Male	31.5	1867	Klebsiella pneumoniae	HAI	R	R	R	S
2022-10-21	P0101	Chris Hani Baragwanath	63	Male	33.7	1956	Group B Streptococcus	HAI	I	S	S	S
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2021-03-24	P0108	Inkosi Albert Luthuli	11	Male	35.9	2008	Staphylococcus aureus	HAI	S	S	R	S
2022-04-05	P0123	Chris Hani Baragwanath	80	Female	28.8	1371	Staphylococcus aureus	HAI	S	I	R	R
2022-07-12	P0063	Groote Schuur Hospital	57	Male	25.3	854	Group B Streptococcus	HAI	S	S	S	I

Source: Synthetic pathogen-level neonatal sepsis dataset (n = 200; 100 EONS, 100 HAI) generated to illustrate WISCA methodology. Data are simulated and do not represent real patients.

What do I need to consider in preprocessing?

- What are the antibiotic combinations we want to include?

  Ampicillin+Gentamicin, Piptaz+Amikacin, Meropenem+Vancomycin, Meropenem

- Any pathogens we want to exclude?

  Contaminants, Coagulase-negative Staphylococcus*, Unknown name, Candida Spp.

- Is the dataset deduplicated for first isolates?

  Not sure - will specify “no” to be on the safe side

- How would you like to code intermediate susceptibility?

  Choose to code Intermediate as Susceptible

- What do I want to stratify by? (Do I have enough data to do so?)

WISCA preprocessing

Once the data is in it's clean and complete form, run preprocessing:

```
antibiotic_in_demo <- c("meropenem","piptaz+amikacin","meropenem+vancomycin")
include_pathogens <- 10
exclude_path_demo <- c("Coagulase-Negative Staphylococcus", "(Unknown Name)","Candida Spp.")
deduped <- "no"
susceptible_I_in <- "S"

source("./Code/wisca_preprocess_amr.R")
```

Under the hood??

Summary: WISCA Preprocessing

```
data_wisca_demo <- wisca_preprocess(x = data_in,
                                    antibiotic_in = antibiotic_in_demo,
                                    exclude = exclude_path_demo,
                                    susceptible_I = susceptible_I_in,
                                    isolate_first = deduped)
```

-Antimicrobial coding
-Microorganism coding
-Organism exclusions and grouping
-Susceptibility coding
-First isolates
-Variable naming

Data needed for WISCA function:

```
antibiotic_wisca_demo <- data_wisca_demo[[1]]
data_wisca_in_demo <- data_wisca_demo[[2]]
```

Antimicrobials
Processed data

Hands-On Part 2: Running the WISCA Function

All on GitHub!

WISCA_training Public

Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file | Add file | Code

BoltonL update code — 4df5faf · 1 hour ago — 15 Commits

Data	update data and load new scripts	3 days ago
output	update	1 hour ago
scripts	update code	1 hour ago
slides	add	3 days ago
workbook	add	3 days ago
.gitignore	first output	2 hours ago
LICENSE	Initial commit	last month
README.md	add	3 days ago
WISCA_training.Rproj	set up script	3 days ago

README | MIT license

From Data to Decision: Applying WISCA in Sepsis Using R

Authors: Larisse Bolton, Matthijs Barends, Aislinn Cook

Welcome! This repository contains all materials for the live training. This training introduces the **Weighted Incidence Syndromic Combination Antibiogram (WISCA)** approach and demonstrates how to prepare data, run the WISCA function, and interpret outputs in R.

What You Will Learn

By the end of this webinar, you will be able to:

About

Training materials for WISCA implementation

Readme
MIT license
Activity
0 stars
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Report repository

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 2

msberends Matthijs Berends
BoltonL

Languages

R 100.0%

AMR for R Assistant

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chat.amr-for-r.org

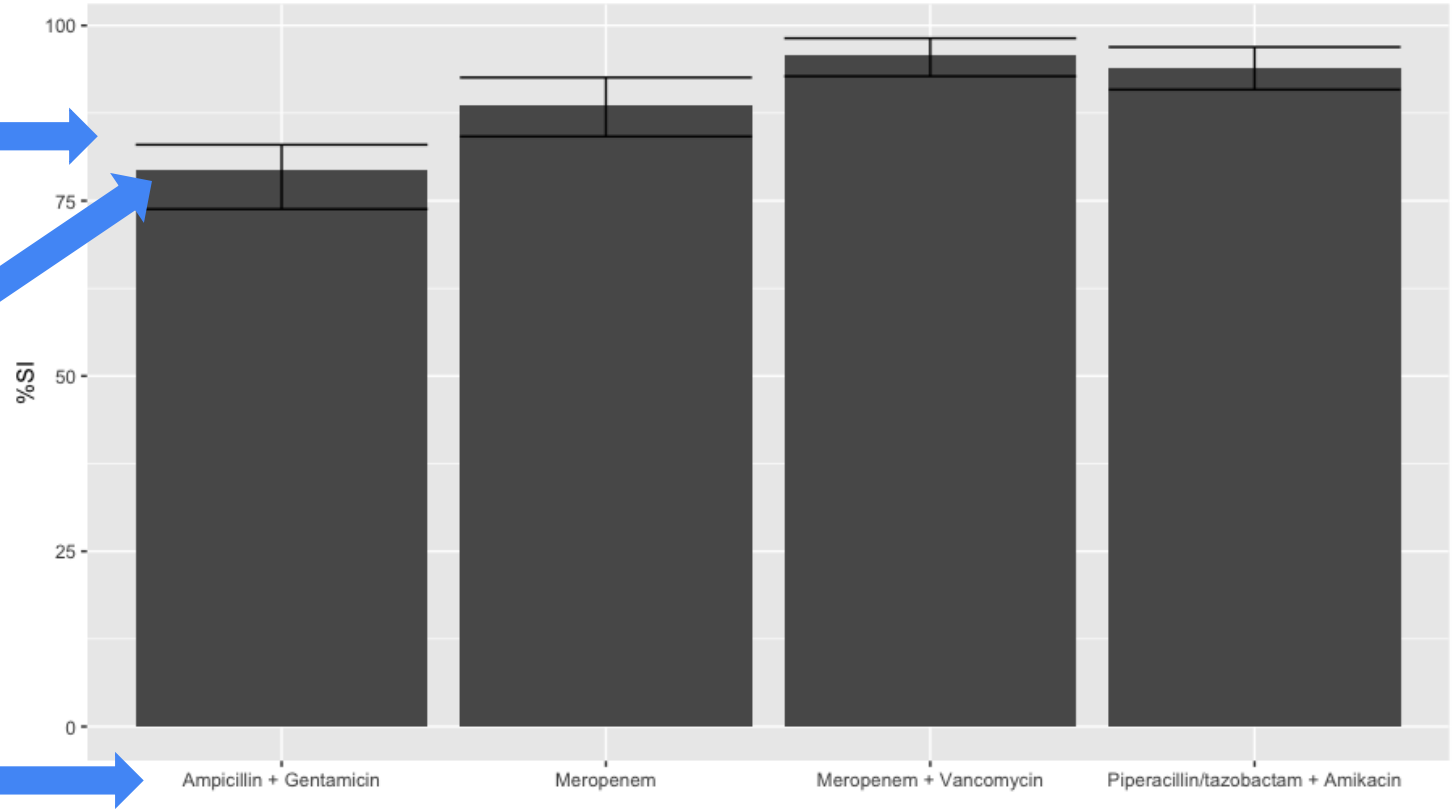
Interpreting outputs

Overall

Overlapping CIs

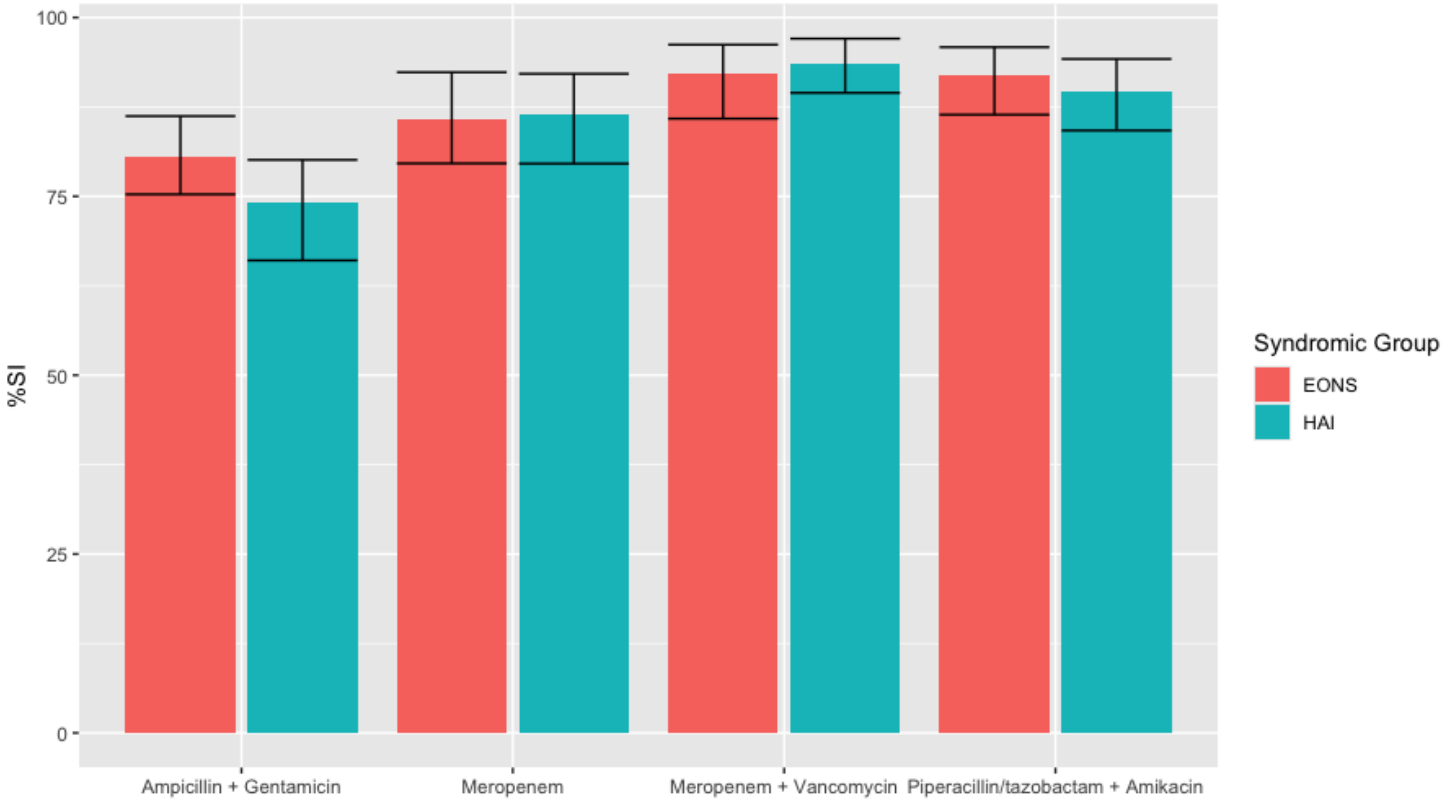
95% credible interval (uncertainty)

Point estimate



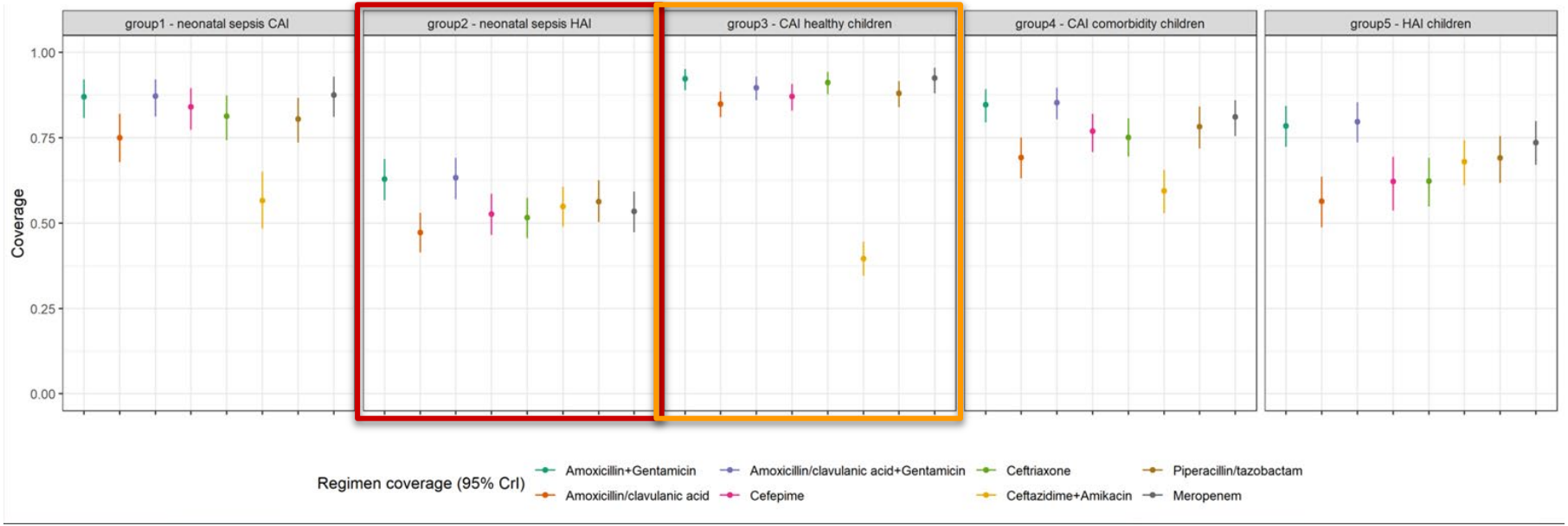
Regimens

Stratified by syndrome



Syndrome stratification

By clinical risk group

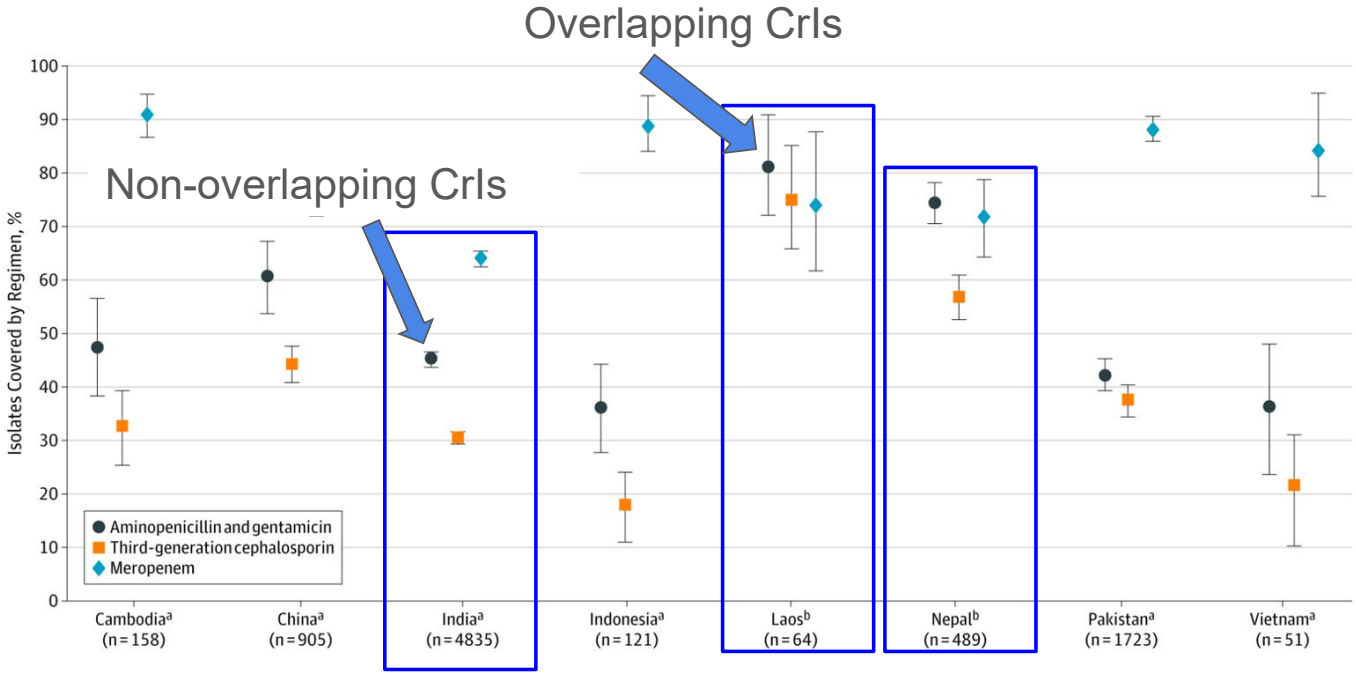


Important differences in coverage by clinical risk groups

Cook et al., 2023 PMID: 37252038

Non-inferiority principles

Is the narrower spectrum regimen within the same bounds as the wider spectrum one?



Bielicki et al., 2020, PMID: 32049298

Table format of outputs

syndromic_group	ab	coverage	lower_ci	upper_ci	n_total	n_tested	n_susceptible
EONS	Ampicillin + Gentamicin	0.811087	0.729253	0.864811	100	100	86
EONS	Piperacillin/tazobactam + Amikacin	0.916735	0.853174	0.958796	100	100	98
EONS	Vancomycin + Meropenem	0.923448	0.873904	0.959981	100	100	99
EONS	Meropenem	0.847764	0.784285	0.90453	100	100	91
HAI	Ampicillin + Gentamicin	0.735553	0.655141	0.799642	100	100	78
HAI	Piperacillin/tazobactam + Amikacin	0.896725	0.830225	0.932542	100	100	96
HAI	Vancomycin + Meropenem	0.929678	0.884944	0.97064	100	100	100
HAI	Meropenem	0.862816	0.794211	0.917295	100	100	92

Pip-taz + amik slightly higher coverage than Amp + gent for both EOS and HAI

Coverage of amp+gen lower in HAI but overlapping CrIs

These choices may matter for interpretation***

You are still in the driver's seat even with a more automated tool

- What are the antibiotic combinations we want to include?

  Ampicillin+Gentamicin, Piptaz+Amikacin, Meropenem+Vancomycin, Meropenem

- Any pathogens we want to exclude?

  Contaminants, Coagulase-negative Staphylococcus*, Unknown name, Candida Spp.

- Is the dataset deduplicated for first isolates?

  Not sure - will specify "no" to be on the safe side

- How would you like to code intermediate susceptibility?

  Choose to code Intermediate as Susceptible

- What do I want to stratify by? (Do I have enough data to do so?)

Syndromic - in the name

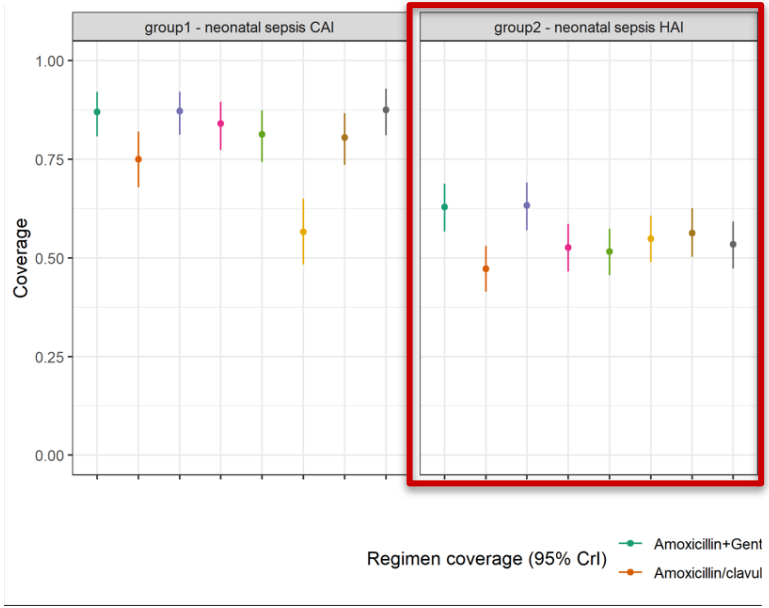
Other things you can stratify by:

- Syndrome
- Risk groups
- Ward
- Age groups

Need to be mindful of small numbers

- **No consensus on optimal threshold for coverage - this will vary by syndrome and hospital!**

- You're comparing coverage **<u>within</u>** the syndrome of choice
  - Coverage overall might be lower than you want but you're trying to pick the best regimen of the ones you can for those patients



Threshold for updating guidelines

- No consensus - research questions still to answer; will depend on local data quality & outputs

- To consider: time of monitoring, sample size, sampling bias before updating guidelines

Wrap-up: Integration and Limitations

Integration into workflows

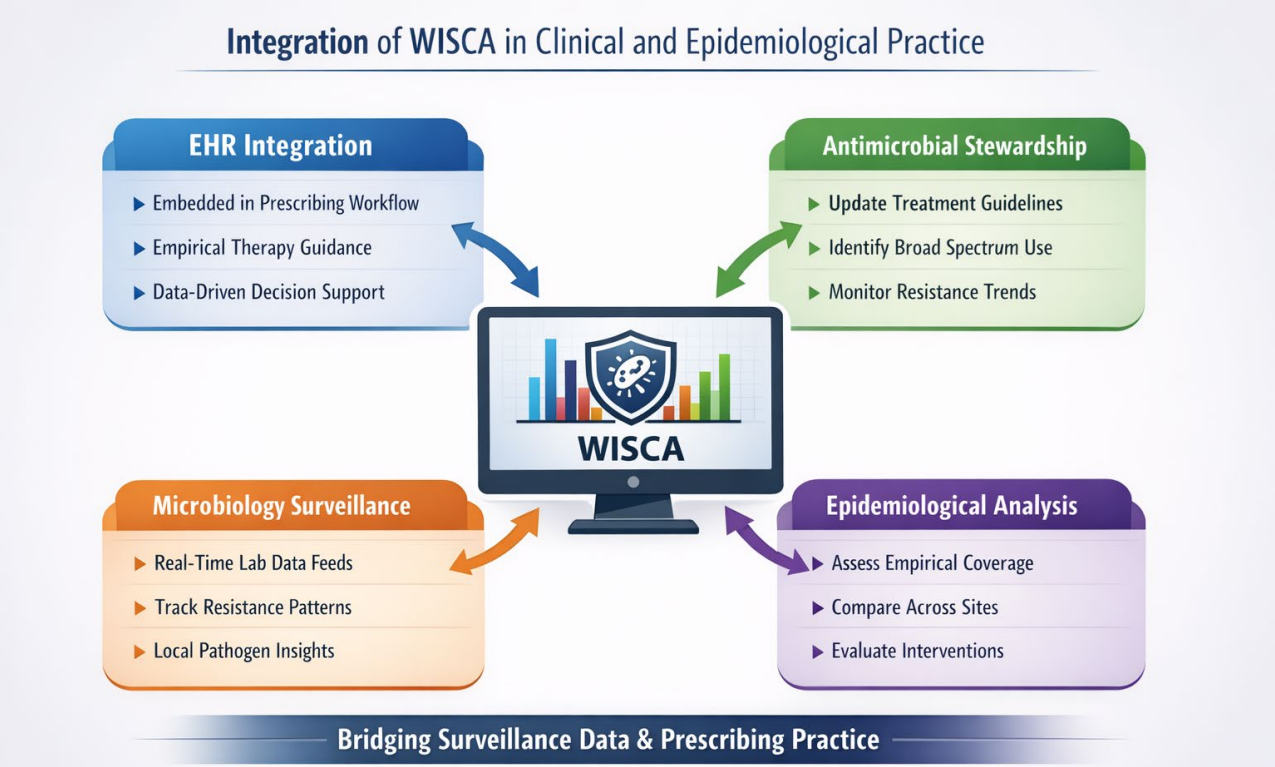


Figure created with assistance from ChatGPT (OpenAI), 2026.

Limitations

Figure created with assistance from ChatGPT (OpenAI), 2026.

Q & A

Resources

- **WISCA Vignette:**

  https://amr-for-r.org/articles/WISCA.html

- **R for data science:**

  https://r4ds.hadley.nz/

- **The Epidemiologist R handbook:**

  https://www.epirhandbook.com/en/

Further reading

- Hebert C, Ridgway J, Vekhter B, Brown EC, Weber SG, Robicsek A. Demonstration of the Weighted-Incidence Syndromic Combination Antibiogram: An Empiric Prescribing Decision Aid. Infect Control Hosp Epidemiol. 2012;33(4):381-388. doi:10.1086/664768
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Feedback

We would love to hear from you so that we can continue to improve the trainings!

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From Data to Decision: Feedback

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Thank you for participating in the WISCA training.